

Project 6

Lecturer: Shuai Wang

TA: Jiawei Liu

Assignment

1. Content

You are required to complete a comprehensive database system, including the implementation of both front-end and back-end processes.

Below are the specific requirements for the content.

- ① **Presentation:** You can construct the **Web system or App** (desktop app, mobile app, WeChat app) using either **B/S or C/S** architecture.
- ② **Framework:** No restrictions. You can utilize **any architecture or framework** that you have researched and are familiar with to implement the front-end and back-end.
- ③ **Database:** No restrictions. You can use databases such as **Access, MySQL, or others** as the storage medium for data.
- ④ **Task Type:** Team work. **3~4 students per group.**

Assignment

1. Content

- ⑤ **Scenario:** You can use real-world database scenarios as the context for this experiment, including but not limited to the given example: **Internet Bookstore** .

Note: **Internet Bookstore** is a classic example, providing a reference and defining the scope of the task. But it also has some mandatory:

- **Internet Bookstore** is the most basic requirement. The **complexity and difficulty of the database scenario** you customize should **not be lower than this example**.
- **Internet Bookstore** is not the minimum standard, and choosing it **does not mean a lower score**. We provide this example to guide everyone in understanding the scope and process of the required task. There is **no good or bad in the database scenario**; everything depends on your effort and completeness.

Assignment

1. Content

⑥ Function Requirements:

Implement possible functions and interfaces for the scenario you have defined. The interface will be presented as web pages for the Web system ,or as new windows or UI in the Apps.

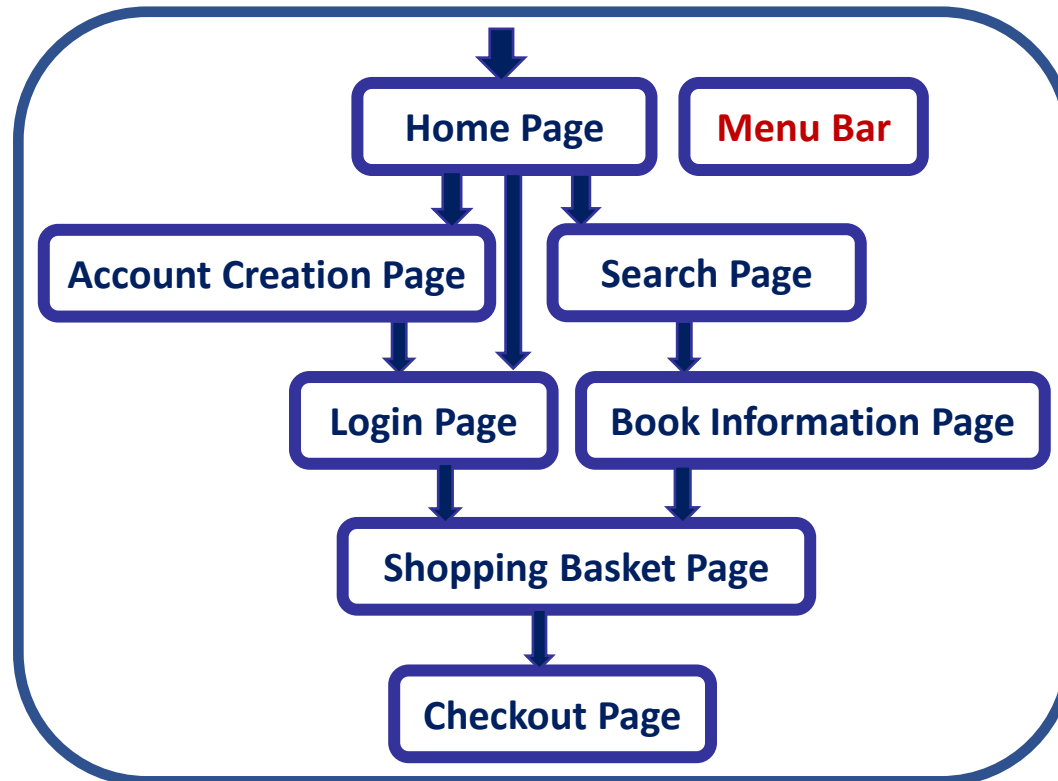
To provide a unified technical path for everyone to experiment with, we take **Internet Bookstore** as an example. The following pages must be implemented (**All pages needed**):

- I. Home Page
- II. Search Page
- III. Book Information Page
- IV. Account Creation Page
- V. Shopping Basket Page
- VI. Login Page
- VII. Checkout Page

Assignment

1. Content

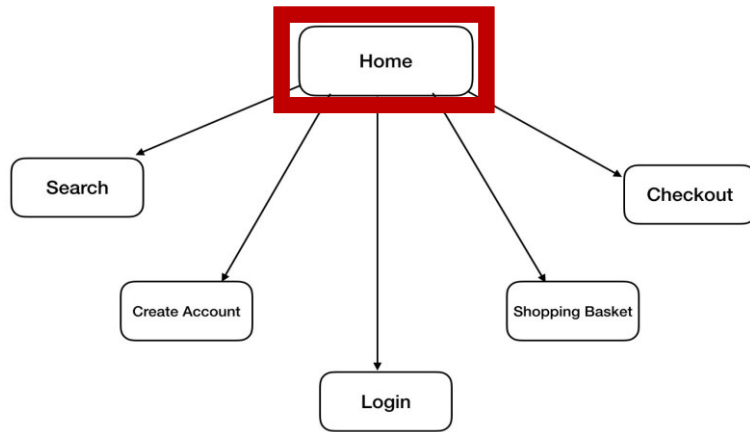
- ⑦ **Function Framework** : The relationship between the above pages is as follows. The following relationships are provided as a reference only. You can design a more flexible framework and presentation method.



Assignment

2. Pages Requirements

- ① **Home Page:** The home page will be the first page users see when they connect to the website or use the application. Users will be greeted by a welcome message for our web site or application, or they can quickly understand what kind of system this is.



Assignment

2. Pages Requirements

- ① **Home Page:** We suggest every page will have a **menu bar** at the top (other methods to display navigation information is ok, such as sidebars, tables of contents, etc.) which allow users to move fluidly around our system from any page.



Assignment

2. Pages Requirements

① Home Page



Assignment

2. Pages Requirements

① Home Page



Assignment

2. Pages Requirements

① **Home Page:** They will also be presented with the following list of options:

Action	Directed to
Creat new Account	Page-Account Creation
Login	Page-Login



Assignment

2. Pages Requirements

② **Search Page:** This page will consist of a series of questions that will assist a user in searching our database. A User selects whether they would like to **search for a book by author, title, or ISBN number**. They then enter a non-case sensitive text string used to search for their book. Some examples of possible queries are:

- Author = Charles Dickens
- Title = American Pastoral
- ISBN = 1829853333

Users will **be presented with a list of books** and have the **ability to view individual books book information pages**, Page Book Information, for each book in the database.

Assignment

2. Pages Requirements

② Search Page



The screenshot shows the Dangdang.com homepage. At the top, there's a search bar with the text "小屁孩日记双语版" and a dropdown menu for "全部分类". Below the search bar, there's a navigation bar with categories like "图书", "米吴科学大侦探", "特装书", "亲签书", "电子书", "课程", "正版教材", "特色书城", and "童装童鞋". A large banner in the center reads "好书礼遇季 图书5折封顶". On the left, there's a sidebar with "图书分类" and various sub-categories like "特色书单", "电子书", "教育", "小说", "文艺", "青春文学", "童书", "人文社科", and "经管". At the bottom, there are several book covers displayed.



The screenshot shows the CNKI (China National Knowledge Infrastructure) search interface. At the top, there's a navigation bar with "cnki 中国知网", "总库", "检索", "CNKI AI", "出版来源", and "我的CNKI". Below the navigation bar, there's a search bar with the text "高级检索". The main search area has a sidebar with "文献分类" and a search form with fields for "主题", "作者", and "文献来源". There are also checkboxes for "OA出版", "网络首发", "增强出版", "基金文献", "中英文扩展", and "同义词扩展". At the bottom, there's a "时间范围" section with "发表时间" and "更新时间" options, and a "检索" button.

Assignment

2. Pages Requirements

- ③ **Book Information Page:** For each book in system, users will be presented with information such as the books title, author, and price.

图书 > 小说 > 科幻小说 > 三体全3册套装三体123黑暗森林死神永生: 刘慈欣代表作, 亚洲“雨



当当自营 三体全3册套装三体123黑暗森林死神永生: 刘慈欣代表作, 亚洲“雨果奖”获奖作品! 中国科幻基石丛书 科幻小说代表作

《三体》第73届世界科幻雨果奖获奖作品, 银河奖特别奖, 《三体3》轨迹奖长篇科幻小说! 2017年世界雨果奖提名作品。跨山海刷边亲签版 下单戳>>

作者: 刘慈欣 出版社: 重庆出版社 出版时间: 2010年11月
在当当小说畅销榜排名11位 ★★★★★ 3967890条评论

抢购价 降价通知
¥46.50 (5折)
定价¥93.00

领券 209减20 109减10

促销 限时抢 时间: 2025.12.08 00:00:00-2025.12.20 23:59:59, 抓紧抢购 详情>>
① 部分促销不可共享, 请以购物车能享受的促销为准

配送至 北京市昌平区城北街道 有货 运费6元, 满49元包邮

服务 由“当当”发货, 并提供售后服务。明日18:55前完成下单, 预计后天(12月19日)可送达

关联商品

新版三体全3册	新版三体全3册	新版三体1	新版三体2	新版三体3
三体全3册	三体刘慈欣科幻作品典藏套装	三体精装纪念版	三体1	
三体2黑暗森林	三体3死神永生	三体典藏版123全集	三体1(典藏版)	
三体2黑暗森林(典藏版)	三体3死神永生(典藏版)			

系列

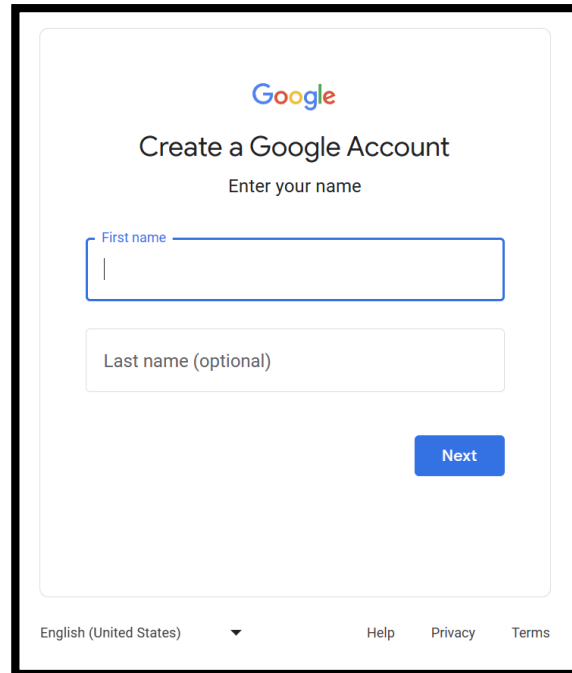
行歌三叠	三体全3册	三体刘慈欣科幻作品典藏套装	三体精装纪念版	三体1
三体2黑暗森林	三体3死神永生	新版三体全3册	新版三体3册套	新版三体1
新版三体2	新版三体3	超新星纪元(新版)	球状闪电(新版)	

1 + 加入购物车 立即购买

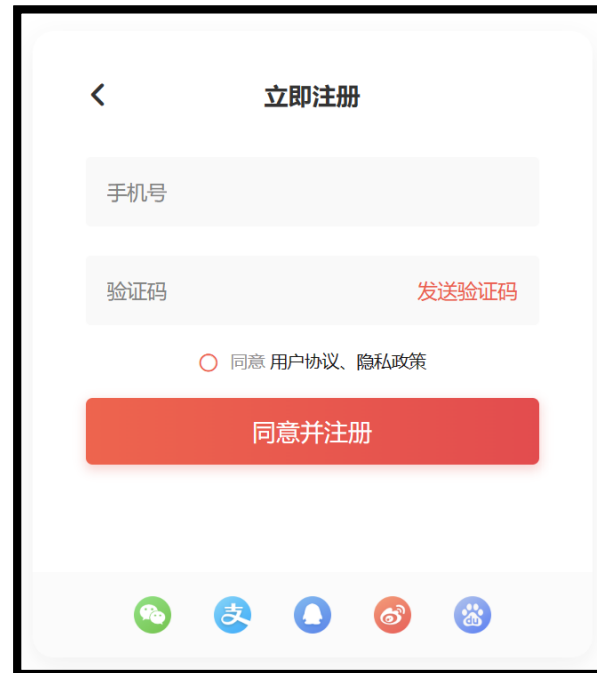
Assignment

2. Pages Requirements

- ④ **Account Creation Page:** Users without accounts or those wishing to create new ones can do so by choosing to create a new account from the system Page Home. They will be asked a series of personal questions including choosing a new **username** for the system as well as a **password**. If any of the information they provide is **incorrect, missing, or invalid**, they will be asked to **enter it again**.



The screenshot shows the Google Account creation interface. At the top is the Google logo, followed by the text "Create a Google Account" and "Enter your name". There are two input fields: "First name" (with a blue border and a cursor) and "Last name (optional)". A blue "Next" button is at the bottom right. At the very bottom, there is a language selector set to "English (United States)" and links for "Help", "Privacy", and "Terms".



The screenshot shows a Chinese version of an account creation page. At the top is a back arrow and the text "立即注册" (Register Now). There are two input fields: "手机号" (Mobile Number) and "验证码" (Verification Code). To the right of the verification code field is a red link "发送验证码" (Send Verification Code). Below these fields is a radio button labeled "同意 用户协议、隐私政策" (Agree to Terms of Service, Privacy Policy). At the bottom is a large red button labeled "同意并注册" (Agree and Register). At the very bottom, there are five circular icons for various services.

- Character count limit
- Character type restriction
- Uniqueness constraint

Assignment

2. Pages Requirements

- ④ **Account Creation Page:** The application layer will check the information the user enters thoroughly each time they submit it before attempting to create the account. If all the data passes, the server will attempt to make the account and check for account creation errors.

After the account information has been entered correctly, the account will be created by the system, and the user will be **automatically logged in** and **directed to Page Shopping Basket** (or Page Search, **decided by you**).

Usually it is directed to the page before login, and the page is in login status. Anyway, it is **decided by you**.

Assignment

2. Pages Requirements

- ⑤ **Shopping Basket Page:** The shopping basket is **a temporary state page** that shows the user which items they have added to their order during their current visit. In addition, users can change their shopping cart by both **modifying the quantities of books** in their baskets, as well as **deleting books** from their basket.

Note: No information contained in the shopping basket is maintained by the system permanently. So a user's **shopping basket is erased each time they connect to our server.**

Assignment

2. Pages Requirements

- ⑥ **Login Page:** The login page will ask a user for their username and password to use to log into the system. In addition, they will have the following options based on the information they provided:

Action	Directed to
Forgot Password	E-mail Message to Administrator
Login: Incorrect Information	Login page
Login: Correct Information	Search page

The application layer will query the database with the information the user has provided and if it is valid for a user in the system, it will log a user in by creating a cookie on the user's machine.

Assignment

2. Pages Requirements

- ⑦ **Checkout Page** : Only when a user has **at least one item in their shopping basket** and has **logged into our system** will they be able to **proceed to the checkout page**.
- First, they will be asked to **confirm the items** in their order.
 - Next, they will be asked to **provide a valid credit card** for the order as well as shown their **shipping address** for the order.
 - Finally, they will be **shown the information** for their entire order and **asked to confirm it**.
 - Once they confirm the order, it is sent to the database for processing.
 - When the **quantity of the items is insufficient**, the **operation will fail** and the **corresponding prompt message will be returned**.
 - If the operation is **successful**, the quantities of the books in stock will be **decremented** by the quantities of books the user purchased for each book.

Assignment

3. Document requirements

- ✓ Hand in a **report** about the TEAM PROJECT (**Each group only needs to submit one report**) :
 - Information of Team Members and Responsibilities
 - Database Scenario
 - Requirements Analysis
 - ER Diagram Design
 - Technical Framework
 - Functional Design & Software Implementation
 - Results Display
- ✓ **code.zip**: Source code and README
- ✓ **PPT/Video/Website Link/Executable Program Package**: The medium of presenting the project. You can choose to use a **PPT** to showcase the website or program, or dynamically showcase the content through a recorded **video**. You can even present the final project as an accessible **website link** or **executable program package**.

Requirements for submitting the assignment

- Email address : liujoy@seu.edu.cn
- File name format : StudentNumber-Name-Project-X

example : *71121000-王五-Project-6.zip*

The team leader's student ID and name.
Each group only needs to submit one.

- The **zip** includes 3 files:
 - ① *report_A6.doc / report_A6.docx / report_A6.pdf*
 - ② *code.zip*
 - ③ *PPT/Video/Website Link/Executable Program Package*

Deadline: Beijing time, 01.22, 23:59:59